

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275342

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

OZEKI; Kenji

LIGHT-EMITTING DEVICE AND METHOD FOR MANUFACTURING LIGHT-EMITTING DEVICE

Abstract

A method for manufacturing a light-emitting device includes providing a structure body. The structure body includes a wiring substrate, a light-emitting element, and a protection element. The wiring substrate includes a wiring part at an upper surface of the wiring substrate. The light-emitting element includes a pair of first electrodes, a first semiconductor layer, and a first element substrate. The protection element includes a pair of second electrodes, a second semiconductor layer, and a second element substrate. The pair of first electrodes and the pair of second electrodes face the upper surface of the wiring substrate. The pair of first electrodes and the pair of second electrodes are connected to the wiring part. The method includes removing at least a portion of the first element substrate of the light-emitting element and at least a portion of the second element substrate of the protection element from the structure body.

Inventors: OZEKI; Kenji (Tokushima-shi, JP)

Applicant: NICHIA CORPORATION (Anan-shi, JP)

Family ID: 1000008434068

Appl. No.: 19/036940

Filed: January 24, 2025

Foreign Application Priority Data

JP	2024-025815	Feb. 22, 2024
JP	2024-182094	Oct. 17, 2024

Publication Classification

Int. Cl.: H10H29/853 (20250101); H10H29/01 (20250101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-025815, filed on Feb. 22, 2024, and the prior Japanese Patent Application No. 2024-182094, filed on Oct. 17, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a light-emitting device and a method for manufacturing a light-emitting device.

BACKGROUND

[0003] In recent years, light-emitting devices that use light-emitting diodes (LEDs) are becoming increasingly popular. It is desirable to increase the light extraction efficiency of such a light-emitting device to realize both higher brightness and power conservation.

SUMMARY

[0004] Embodiments are directed to a light-emitting device in which the light extraction efficiency can be increased and a method for manufacturing such a light-emitting device.

[0005] According to one aspect of the present disclosure, a method for manufacturing a light-emitting device includes: providing a structure body, the structure body including a wiring substrate including a wiring part at an upper surface, a light-emitting element in which a pair of first electrodes, a first semiconductor layer, and a first element substrate are stacked, and a protection element in which a pair of second electrodes, a second semiconductor layer, and a second element substrate are stacked, wherein the pair of first electrodes and the pair of second electrodes face the upper surface of the wiring substrate, and the pair of first electrodes and the pair of second electrodes are connected to the wiring part; and removing at least a portion of the first element substrate of the light-emitting element and at least a portion of the second element substrate of the protection element from the structure body.

[0006] According to one aspect of the present disclosure, a light-emitting device includes a wiring substrate, a light-emitting element arranged on the wiring substrate, a protection element arranged on the wiring substrate, a light-transmitting member arranged on the light-emitting element, and a covering member arranged on the wiring substrate, the covering member not covering an upper surface of the light-transmitting member while covering a lateral surface of the light-emitting element and a lateral surface of the protection element. An upper surface of the light-emitting element and an upper surface of the protection element are positioned in a same plane.

[0007] According to embodiments, a light-emitting device in which the light extraction efficiency can be increased and a method for manufacturing such a light-emitting device can be realized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIGS. 1 to 4 are cross-sectional views showing a method for manufacturing a light-emitting device according to a first embodiment;

[0009] FIG. 5 is a cross-sectional view showing the light-emitting device according to the first embodiment;

[0010] FIG. 6 is a perspective view showing the light-emitting device according to the first

embodiment;

[0011] FIGS. **7** and **8** are cross-sectional views showing a method for manufacturing a light-emitting device according to a second embodiment;

[0012] FIG. **9** is a cross-sectional view showing the light-emitting device according to the second embodiment;

[0013] FIG. **10** is a cross-sectional view showing a light-emitting device according to a third embodiment;

[0014] FIG. **11** is a cross-sectional view showing a method for manufacturing a light-emitting device according to a fourth embodiment;

[0015] FIG. **12** is a cross-sectional view showing the light-emitting device according to the fourth embodiment;

[0016] FIG. **13** is a perspective view showing a light-emitting device according to a fifth embodiment;

[0017] FIG. **14** is a cross-sectional view along line XIV-XIV shown in FIG. **13**;

[0018] FIG. **15** is a cross-sectional view along line XV-XV shown in FIG. **13**;

[0019] FIG. **16A**, FIG. **16B**, and FIGS. **17A** to **17C** are micrographs showing appearances after polishing a light-emitting element, a protection element, and a first covering member;

[0020] FIG. **18** is a table showing measurement results of surface roughnesses of a light-emitting element, a protection element, and a first covering member; and

[0021] FIG. **19** is a graph showing a surface profile of one sample, the horizontal axis is a horizontal-direction position; and the vertical axis is the vertical-direction position.

DETAILED DESCRIPTION

[0022] A method for manufacturing a light-emitting device and a light-emitting device according to embodiments will now be described with reference to the drawings. The drawings are schematic and conceptual, and are enhanced and simplified as appropriate. Therefore, the dimensions of the components are not necessarily an accurate representation of the actual product. The dimensional ratios, numbers, positional relationships, etc., of the components do not necessarily match between the drawings. This is similar for the other drawings described below as well. The sizes, positional relationships, etc., of the members shown in the drawings may be exaggerated for clarity of description. Furthermore, to avoid excessive complexity of the drawings, schematic views may be used in which some of the components are not illustrated; and end views may be used in which only a cross section is shown as a cross-sectional view. According to the embodiment, “covered” and “cover” are not limited to direct contact, and include the case of covering and being covered indirectly via, for example, another member.

First Embodiment

[0023] FIGS. **1** to **4** are cross-sectional views showing a method for manufacturing a light-emitting device according to the embodiment.

[0024] FIG. **5** is a cross-sectional view showing the light-emitting device according to the embodiment.

[0025] FIG. **6** is a perspective view showing the light-emitting device according to the embodiment.

Process of Providing Structure Body **100**

[0026] As shown in FIG. **1**, a wiring substrate **10**, a light-emitting element **20**, and a protection element **30** are provided. The order in which these components are provided is arbitrary.

[0027] The wiring substrate **10** includes an insulating base body **11** and a conductive wiring part **12**. The wiring part **12** is located at least at an upper surface **10a** of the wiring substrate **10**. In addition to the upper surface **10a** of the wiring substrate **10**, the wiring part **12** may be located inside the wiring substrate **10** and at a lower surface **10b** of the wiring substrate **10**.

[0028] The light-emitting element **20** is, for example, a light-emitting diode. The light-emitting element **20** includes a pair of first electrodes **21**, a first semiconductor layer **22**, and a first element

substrate **23**. In the light-emitting element **20**, the pair of first electrodes **21**, the first semiconductor layer **22**, and the first element substrate **23** are stacked in this order. The first element substrate **23** is, for example, a growth substrate of the first semiconductor layer **22** and is, for example, a sapphire substrate.

[0029] In the process of providing a structure body **100** (hereinbelow, also called “the provision process”), considering ease of mounting to the wiring substrate **10**, the thickness of the light-emitting element **20** to be provided is not less than 100 μm , e.g., about 150 μm . Within the thickness of the light-emitting element **20**, the thickness of the first element substrate **23** is the thickness of the light-emitting element **20** minus the thickness of the first electrode **21** and the thickness of the first semiconductor layer **22**, and is, for example, about 130 μm . It is sufficient for the light-emitting element **20** to include at least one positive and negative pair of first electrodes **21** at the same surface side of the light-emitting element **20**; and multiple pairs of first electrodes **21** or different numbers of positive and negative first electrodes **21** may be provided at the same surface side of the light-emitting element **20**. The thickness of the first electrode **21** is, for example, about 10 μm .

[0030] The first semiconductor layer **22** includes, for example, a gallium nitride (GaN) semiconductor. The thickness of the first semiconductor layer **22** is, for example, about 8 μm to 10 μm . In the first semiconductor layer **22**, an n-type semiconductor layer **24**, an active layer **25**, and a p-type semiconductor layer **26** are stacked in this order from the first element substrate **23** side. Among the pair of first electrodes **21**, one of the first electrodes **21** is connected to the n-type semiconductor layer **24**; and the other of the first electrodes **21** is connected to the p-type semiconductor layer **26**. In the specification, “connect” means an electrical connection.

[0031] The protection element **30** protects the light-emitting element **20** from a surge current or static electricity and is, for example, a Zener diode. The protection element **30** includes a pair of second electrodes **31**, a second semiconductor layer **32**, and a second element substrate **33**. In the protection element **30**, the pair of second electrodes **31**, the second semiconductor layer **32**, and the second element substrate **33** are stacked in this order. The second element substrate **33** is, for example, a semiconductor substrate and is, for example, a silicon (Si) substrate. In the provision process, considering ease of mounting to the wiring substrate **10**, the thickness of the protection element **30** to be provided is not less than 100 μm , e.g., about 150 μm . The thickness of the second element substrate is, for example, about 120 μm . It is sufficient for the protection element **30** to include at least a positive and negative pair of second electrodes **31** at the same surface side of the protection element **30**; and multiple pairs of second electrodes **31** or different numbers of positive and negative second electrodes **31** may be provided at the same surface side of the protection element **30**. The thickness of the second electrode is, for example, about 10 μm .

[0032] The second semiconductor layer **32** includes, for example, a silicon semiconductor. The thickness of the second semiconductor layer **32** is, for example, about 20 μm . The second semiconductor layer **32** includes, for example, a base material layer **34** of the p-conductivity type, a first semiconductor region **35** of the p.sup.+ -conductivity type, and a second semiconductor region **36** of the n.sup.+ -conductivity type. The impurity concentration of the first semiconductor region **35** is greater than the impurity concentration of the base material layer **34**. The first semiconductor region **35** and the second semiconductor region **36** are exposed at the lower surface of the base material layer **34**. The second semiconductor region **36** is separated from the first semiconductor region **35** with the base material layer **34** interposed. Among the pair of second electrodes **31**, one of the second electrodes **31** is connected to the first semiconductor region **35**; and the other of the second electrodes **31** is connected to the second semiconductor region **36**.

[0033] The light-emitting element **20** and the protection element **30** are disposed on the wiring substrate **10** after providing the wiring substrate **10**, the light-emitting element **20**, and the protection element **30**. At this time, it is preferable for the light-emitting element **20** and the protection element **30** to have a flip-chip connection to the wiring substrate **10**. In other words, the

first electrodes **21** of the light-emitting element **20** are caused to face the wiring parts **12** of the wiring substrate **10**; and the first electrodes **21** are bonded to the wiring parts **12**. The second electrodes **31** of the protection element **30** are caused to face the wiring parts **12** of the wiring substrate **10**; and the second electrodes **31** are bonded to the wiring parts **12**. The order of the process of disposing the light-emitting element **20** on the wiring substrate **10** and the process of disposing the protection element **30** on the wiring substrate **10** is arbitrary.

[0034] The first electrodes **21** of the light-emitting element **20** and/or the second electrodes **31** of the protection element **30** can be connected with the wiring parts **12** of the wiring substrate **10** via a conductive bonding member. Examples of the bonding member include known bonding members such as eutectic solder, conductive paste, bumps, plating, etc. The first electrodes **21** of the light-emitting element **20** and/or the second electrodes **31** of the protection element **30** may be directly bonded with the wiring parts **12** of the wiring substrate **10** without a bonding member interposed.

[0035] Thus, the structure body **100** is provided. The structure body **100** includes the wiring substrate **10**, the light-emitting element **20**, and the protection element **30**. On one wiring substrate **10** in the structure body **100**, only one set of at least one light-emitting element **20** and at least one protection element **30** for protecting the at least one light-emitting element **20** may be included, or multiple sets may be included. FIGS. **1** to **4** show only one set of one light-emitting element **20** and one protection element **30**. In the structure body **100**, a pair of first electrodes **21** of the light-emitting element **20** and a pair of second electrodes **31** of the protection element **30** face the upper surface **10a** of the wiring substrate **10** and are connected to the wiring parts **12**.

[0036] The structure body **100** may be provided by manufacturing as described above, or may be provided by acquiring externally by procurement, etc.

Process of Disposing First Covering Member **41**

[0037] Then, as shown in FIG. **2**, a first covering member **41** is disposed on the wiring substrate **10**. The first covering member **41** can include, for example, a resin. Examples of the resin used for the first covering member **41** include a silicone resin. The first covering member **41** fixes the first electrodes **21** of the light-emitting element **20** and the wiring parts **12** of the wiring substrate **10**, and fixes the second electrodes **31** of the protection element **30** and the wiring parts **12** of the wiring substrate **10**. For example, the first covering member **41** can cover the entire light-emitting element **20** and the entire protection element **30**. In other words, the first covering member **41** covers the first element substrate **23** of the light-emitting element **20** and the second element substrate **33** of the protection element **30**. The first covering member **41** may be located between the wiring substrate **10** and the light-emitting element **20** and between the wiring substrate **10** and the protection element **30**.

[0038] In the process of disposing the first covering member **41**, an uncured resin that is used to form the first covering member **41** is disposed on the wiring substrate **10**. The uncured resin for configuring the first covering member **41** can be disposed by, for example, potting, spraying, printing, transfer molding, injection molding, compression molding, etc. Subsequently, the first covering member **41** is formed by curing the resin. It is preferable for the first covering member **41** after curing to have a lower hardness than the first and second element substrates **23** and **33**. As a result, in the first removal process described below, the first covering member **41** can be removed together with the light-emitting element **20** and the protection element **30**; and cracking and/or chipping of the light-emitting element **20** and/or the protection element **30** can be reduced.

First Removal Process

[0039] Then, as shown in FIG. **3**, at least a portion of the first element substrate **23** of the light-emitting element **20** and at least a portion of the second element substrate **33** of the protection element **30** are removed from the structure body **100**. Examples of the removal method include known methods such as polishing, cutting, etching, blasting, etc. Among such methods, polishing is preferable. According to the embodiment, a portion of the first element substrate **23**, a portion of the second element substrate **33**, and a portion of the first covering member **41** can be removed by

simultaneously polishing the structure body **100** from the surface at the side opposite to the wiring substrate **10**, i.e., the upper surface side of the first covering member **41**. As a result, the upper surface of the first element substrate **23**, the upper surface of the second element substrate **33**, and the upper surface of the first covering member **41** exposed by the polishing can be in the same plane. Also, the height variation of the light-emitting element **20** and the protection element **30** between manufactured light-emitting devices can be reduced.

[0040] As a result, in the structure body **100** in which the first covering member **41** is disposed, the first element substrate **23**, the second element substrate **33**, and the first covering member **41** of the structure body **100** are polished from the upper surface side; and a new upper surface **23a** of the first element substrate **23** and a new upper surface **33a** of the second element substrate **33** are exposed from under the first covering member **41**. The upper surface **23a**, the upper surface **33a**, and an upper surface **41a** of the first covering member **41** of the structure body **100** after polishing are positioned in the same virtual plane **90**. At this time, the first semiconductor layer **22** and the first electrodes **21** of the light-emitting element **20** and the second semiconductor layer **32** and the second electrodes **31** of the protection element **30** remain in the structure body **100** without being removed by the polishing. For example, the thicknesses of the light-emitting element **20**, the protection element **30**, and the first covering member **41** after polishing are set to be not less than 20 μm and not more than 80 μm , e.g., about 60 μm .

Process of Disposing Light-Transmitting Member **50**

[0041] Then, as shown in FIG. **4**, a light-transmitting member **50** is disposed on the light-emitting element **20** of the structure body **100**. The light-transmitting member **50** is a plate-like member including an upper surface **50a**, a lower surface **50b** at the side opposite to the upper surface **50a**, and a lateral surface **50c** between the upper surface **50a** and the lower surface **50b**. The light-transmitting member **50** is disposed on the light-emitting element **20** so that the lower surface **50b** of the light-transmitting member **50** faces an upper surface **20a** of the light-emitting element **20** (here, the upper surface **23a** of the first element substrate **23**). For example, the light-transmitting member **50** can be bonded to the upper surface **23a** of the first element substrate **23** of the light-emitting element **20** by a bonding material layer **51**. The bonding material layer **51** is light-transmissive. The light-transmitting member **50** may be directly bonded to the light-emitting element **20** without the bonding material layer **51** interposed.

[0042] The area of the lower surface **50b** of the light-transmitting member **50** may be equal to the area of the upper surface **20a** of the light-emitting element **20**, greater than the area of the upper surface **20a** of the light-emitting element **20**, or less than the area of the upper surface **20a** of the light-emitting element. It is preferable for the area of the lower surface **50b** of the light-transmitting member **50** to be not less than the area of the upper surface **20a** of the light-emitting element **20**, and more preferably equal to the area of the upper surface **20a** of the light-emitting element **20**.

[0043] By setting the entire upper surface **20a** of the light-emitting element **20** to face the lower surface **50b** of the light-transmitting member **50** in a plan view, greater amount of the light emitted from the light-emitting element **20** can be incident on the lower surface **50b** of the light-transmitting member **50**. Also, by setting the entire lower surface **50b** of the light-transmitting member **50** to face the upper surface **20a** of the light-emitting element **20**, the light that is emitted from the upper surface **20a** of the light-emitting element **20** can be perpendicularly incident on the lower surface **50b** over the entire lower surface **50b** of the light-transmitting member **50**. As a result, uneven brightness and uneven chromaticity at the upper surface **50a**, which is the light extraction surface of the light-transmitting member **50**, can be reduced.

[0044] When the light-transmitting member **50** is located on the light-emitting element **20** with the bonding material layer **51** interposed, the bonding material layer **51** is located in at least the region in which the lower surface **50b** of the light-transmitting member **50** and the upper surface **20a** of the light-emitting element **20** face each other. It is preferable for the bonding material layer **51** to cover the entire upper surface **20a** of the light-emitting element **20** and the entire lower surface **50b**

of the light-transmitting member **50** in a plan view. As a result, greater amount of the light emitted from the light-emitting element **20** can be incident on the lower surface **50b** of the light-transmitting member **50** via the bonding material layer **51**.

[0045] The light-transmitting member **50** includes, for example, a phosphor. The phosphor absorbs the light emitted from the light-emitting element **20** and emits light of a different wavelength. According to the embodiment, the light-transmitting member **50** is a plate-shaped ceramic containing a phosphor, and is, for example, a sintered body of a YAG (yttrium aluminum garnet) phosphor and aluminum oxide (Al.sub.2O.sub.3). However, as described below, the configuration of the light-transmitting member **50** is not limited thereto.

Process of Disposing Second Covering Member **42**

[0046] Then, as shown in FIG. 5, a second covering member **42** is disposed on the first covering member **41**. For example, the second covering member **42** can be disposed using potting, spraying, printing, compression molding using a mold, etc. The second covering member **42** covers an upper surface **30a** of the protection element **30** and the lateral surface **50c** of the light-transmitting member **50** and does not cover the upper surface **50a** of the light-transmitting member **50**. The second covering member **42** is, for example, a resin material and includes, for example, a silicone resin. The second covering member **42** covers the lateral surface **50c** of the light-transmitting member **50** that is included in the light-emitting surface of the light-emitting device **1**. It is therefore preferable for the second covering member **42** to have a higher reflectance for the light emitted from the light-emitting element **20** than the first covering member **41**. As a result, leakage light from the upper surface of the second covering member **42** among the light emitted from the light-emitting element **20** can be reduced.

[0047] Continuing, the wiring substrate **10** and a covering member **40** are singulated into each set including one light-emitting element **20** and one protection element **30**. Thus, the light-emitting device **1** according to the embodiment is manufactured. In the light-emitting device **1**, the first covering member **41** and the second covering member **42** configure the covering member **40** of the light-emitting device **1**. The light-emitting device **1** may have a configuration that does not include the second covering member **42**.

Light-Emitting Device **1**

[0048] A configuration of the light-emitting device **1** will now be described.

[0049] As shown in FIGS. 5 and 6, the light-emitting device **1** includes the wiring substrate **10**, the light-emitting element **20**, the protection element **30**, the covering member **40**, and the light-transmitting member **50**. The light-emitting element **20**, the protection element **30**, and the covering member **40** each are located on the wiring substrate **10**. The light-transmitting member **50** is located on the light-emitting element **20**. The covering member **40** covers a lateral surface **20b** of the light-emitting element **20** and the upper surface **30a** and a lateral surface **30b** of the protection element **30**, but does not cover the upper surface **50a** of the light-transmitting member **50**. The upper surface **20a** of the light-emitting element **20** and the upper surface **30a** of the protection element **30** are positioned in the same plane **90**. In the specification, "in the same plane" means that the allowed variation is such that the difference between the surfaces is within $\pm 15\ \mu\text{m}$.

[0050] The wiring substrate **10** includes the insulating base body **11** and the conductive wiring part **12**. The wiring part **12** is located on at least the upper surface **10a** of the wiring substrate **10**. The wiring part **12** may include an anode pad and a cathode pad as external connection terminals connected outside the light-emitting device **1** at the upper surface **10a** or the lower surface **10b** of the wiring substrate **10**.

[0051] The light-emitting element **20** includes the pair of first electrodes **21**, the first semiconductor layer **22**, and the first element substrate **23**. The first electrodes **21** are connected to the wiring parts **12** of the wiring substrate **10**. The first semiconductor layer **22** includes, for example, a gallium nitride semiconductor. In the first semiconductor layer **22**, the p-type semiconductor layer **26**, the active layer **25**, and the n-type semiconductor layer **24** are provided in

this order from the wiring substrate **10** side. The p-type semiconductor layer **26** and the n-type semiconductor layer **24** are connected respectively to the first electrodes **21** of different polarities. The first element substrate **23** is light-transmissive. The first element substrate **23** is, for example, a sapphire substrate. The lower surface of the first element substrate **23** contacts the n-type semiconductor layer **24**. The upper surface **23a** of the first element substrate **23** forms the upper surface **20a** of the light-emitting element **20**.

[0052] In the light-emitting device **1**, the thickness of the light-emitting element **20** is, for example, not less than 20 μm and not more than 80 μm , e.g., about 60 μm . Within the light-emitting element **20**, for example, the thickness of the first semiconductor layer **22** is about 8 to 10 μm ; and the thickness of the first electrode **21** is about 10 μm . By thinning the light-emitting element **20**, the light that is emitted laterally from the lateral surface of the light-emitting element **20**, passes through the covering member **40**, and leaks outside the light-emitting device **1** can be reduced.

[0053] In the first semiconductor layer **22**, the active layer **25** may have a single quantum well (SQW) structure, or may have a multi-quantum well (MQW) structure including multiple well layers. The first semiconductor layer **22** includes multiple semiconductor layers formed of a nitride semiconductor. The nitride semiconductor all includes compositions of semiconductors of the chemical formula $\text{In}_{\text{sub}.x}\text{Al}_{\text{sub}.y}\text{Ga}_{\text{sub}.1-x-y}\text{N}$ ($0 \leq x$, $0 \leq y$, and $x+y \leq 1$) for which the composition ratios x and y are changed within the ranges respectively. The light emission peak wavelength of the active layer can be selected as appropriate according to the purpose. For example, the active layer is configured to emit visible light or ultraviolet light.

[0054] The first semiconductor layer **22** may include multiple light-emitting parts formed of the n-type semiconductor layer **24**, the active layer **25**, and the p-type semiconductor layer **26**. When the first semiconductor layer **22** includes multiple light-emitting parts, each of the light-emitting parts may include well layers having different light emission peak wavelengths, or may include well layers having the same light emission peak wavelength. The light emission peak wavelength being the same also includes cases where there is variation of about several nm. The combination of the light emission peak wavelengths of the multiple light-emitting parts can be selected as appropriate. For example, when the first semiconductor layer includes two light-emitting parts, examples of combinations of the light emitted by the light-emitting parts include blue light and blue light, green light and green light, red light and red light, ultraviolet light and ultraviolet light, blue light and green light, blue light and red light, green light and red light, etc. For example, when the semiconductor structure body includes three light-emitting parts, examples of the combinations of the light emitted by the light-emitting parts include blue light, green light, and red light. Each light-emitting part may include one or more well layers having different light emission peak wavelengths than the other well layers.

[0055] The light-emitting device **1** includes the protection element **30**. Examples of the protection element **30** include a Zener diode, a varistor, a capacitor, etc. By the light-emitting device **1** including the protection element **30**, a reverse voltage applied to the light-emitting element **20** can be shorted, and forward overvoltage applied to the light-emitting element **20** can be shorted. The protection element **30** is, for example, a bare-chip Zener diode. From the perspective of protecting the light-emitting element **20** from overvoltage, it is preferable for the range of the Zener voltage to be about 16 to 40 V.

[0056] The protection element **30** includes the pair of second electrodes **31**, the second semiconductor layer **32**, and the second element substrate **33**. The second electrodes **31** are connected to the wiring parts **12** of the wiring substrate **10**. The second element substrate **33** is, for example, a semiconductor substrate. The semiconductor substrate may be an intrinsic semiconductor that substantially does not include an impurity, or may be a p-type or n-type semiconductor substrate doped with a p-type impurity or an n-type impurity. For example, the second element substrate **33** is a p-type silicon semiconductor substrate.

[0057] The second semiconductor layer **32** is, for example, a silicon semiconductor that includes,

for example, the base material layer **34** of the p-conductivity type and the first semiconductor region **35** and the second semiconductor region **36** arranged to be separated from each other inside the base material layer **34**. The conductivity type of the first semiconductor region **35** is the p.sup.+-type; and the conductivity type of the second semiconductor region **36** is the n.sup.+-type. The first semiconductor region **35** and the second semiconductor region **36** are connected to the second electrodes **31** having respectively different polarities. The lower surface of the second element substrate **33** contacts the base material layer **34**. The upper surface **33a** of the second element substrate **33** forms the upper surface **30a** of the protection element **30**. The thickness of the second semiconductor layer **32** is, for example, about 20 μm .

[0058] In the protection element **30**, the second element substrate **33** and the base material layer **34** of the second semiconductor layer **32** may be formed of different materials, or may be formed of the same material. When the second element substrate **33** and the base material layer **34** of the second semiconductor layer **32** are formed of the same material (e.g., a silicon semiconductor), there are cases where a distinct boundary is not present between the second element substrate **33** and the second semiconductor layer. Even when there is a boundary, there are cases where it is not easy to visually recognize the boundary. In the drawings, the boundary between the second element substrate **33** and the second semiconductor layer **32** is illustrated by a broken line.

[0059] In the light-emitting device **1**, the thickness of the protection element **30** is, for example, about 60 μm . The thickness of the second element substrate **33** is, for example, about 30 μm ; the thickness of the second semiconductor layer **32** is, for example, about 20 μm ; and the thickness of the second electrode **31** is, for example, about 10 μm .

[0060] The covering member **40** is located on the wiring substrate **10**, covers the lateral surface **20b** of the light-emitting element **20** and the lateral surface **30b** of the protection element **30**, and does not cover the upper surface **50a** of the light-transmitting member **50**. The upper surface **30a** of the protection element **30** may be covered with the covering member **40** or may be exposed from the covering member **40**. When the second element substrate **33** is a semiconductor substrate, it is preferable for the upper surface **30a** of the protection element **30** to be covered with the covering member **40**. The covering member **40** is preferably formed of an insulating material. The covering member **40** preferably has light-shielding property, and more preferably light-reflective property. The covering member **40** is, for example, a member in which particles of a light-reflective substance or the like are contained in a light-transmitting resin.

[0061] Examples of the resin included in the covering member **40** include a silicone resin, a modified silicone resin, an epoxy resin, a modified epoxy resin, a polyester resin, a polyimide resin, a modified polyimide resin, etc. It is preferable to use a thermosetting resin having good heat resistance; for example, a silicone resin that has good weather resistance can be used preferably. Examples of the light-reflective substance include titanium oxide, zirconium oxide, boron nitride, and aluminum oxide. The covering member **40** may include an organic material such as a resin or the like, an inorganic material, or both an organic material and an inorganic material. For example, the covering member **40** may include an inorganic material including boron nitride and alkali metal silicate.

[0062] The light-emitting device **1** can include the first covering member **41** and the second covering member **42** as the covering member **40**. The first covering member **41** contacts the wiring substrate **10** and covers the lateral surface **20b** of the light-emitting element **20** and the lateral surface **30b** of the protection element **30**. The upper surface **41a** of the first covering member **41** is positioned in a plane **90**. The second covering member **42** is located on the first covering member **41**. The second covering member **42** covers the upper surface **30a** of the protection element **30** and the lateral surface **50c** of the light-transmitting member **50**. It is preferable for the second covering member **42** to have a higher reflectance for the light emitted from the light-emitting element **20** than the first covering member **41**.

[0063] It is preferable for the surface roughness (Sa) of the upper surface **30a** of the protection

element **30** to be greater than the surface roughness (Sa) of the upper surface **20a** of the light-emitting element **20**. The adhesion between the protection element **30** and the second covering member **42** is improved thereby. Also, it is preferable for the surface roughness (Sa) of the upper surface **41a** of the first covering member **41** to be greater than the surface roughness (Sa) of the upper surface **20a** of the light-emitting element **20** and the surface roughness (Sa) of the upper surface **30a** of the protection element **30**. The adhesion between the first covering member **41** and the second covering member **42** is improved thereby.

[0064] The light-transmitting member **50** is located on the light-emitting element **20**, transmits the light emitted from the light-emitting element **20**, and emits the light externally. For example, the light-transmitting member **50** is bonded to the first element substrate **23** of the light-emitting element **20** via the light-transmitting bonding material layer **51**.

[0065] For example, the light-transmitting member **50** may be formed of any or at least one selected from the group consisting of inorganic material such as a ceramic such as aluminum nitride, aluminum oxide, yttrium oxide, yttrium aluminum perovskite (YAlO₃: YAP), or the like, glass, sapphire, or the like, or an organic material such as a resin including at least one of a silicone resin, a modified silicone resin, an epoxy resin, a modified epoxy resin, an acrylic resin, a phenol resin, a fluororesin, a hybrid resin, etc.

[0066] The light-transmitting member **50** may include a phosphor configured to convert at least a portion of the light emitted by the light-emitting element **20** into light of a different wavelength. Examples of the light-transmitting member **50** containing the phosphor include a sintered body of a phosphor, and a member including a light-transmitting base material such as a resin, a glass, a ceramic, etc., in which phosphor powder is contained. The sintered body of the phosphor may be formed by sintering only a phosphor, or may be formed by sintering a mixture of a phosphor and a substance (e.g., a light-diffusing substance, a sintering aid, etc.) other than a phosphor. The light-transmitting member **50** may include a member in which a phosphor layer is located at the surface of a light-transmitting base body such as a resin, glass, a ceramic, etc.

[0067] Examples of the phosphor include oxynitride-based phosphors such as an yttrium-aluminum-garnet-based phosphor (e.g., (Y, Gd)₃(Al, Ga)₅O₁₂:Ce), a lutetium-aluminum-garnet-based phosphor (e.g., Lu₃(Al, Ga)₅O₁₂:Ce), a terbium-aluminum-garnet-based phosphor (e.g., Tb₃(Al, Ga)₅O₁₂:Ce), a CCA-based phosphor (e.g., Ca₁₀(PO₄)₆Cl₂:Eu), an SAE-based phosphor (e.g., Sr₄Al₁₄O₂₅:Eu), a chlorosilicate-based phosphor (e.g., Ca₈MgSi₄O₁₆Cl₂:Eu), a silicate-based phosphor (e.g., (Ba, Sr, Ca, Mg)₂SiO₄:Eu), a β -sialon-based phosphor (e.g., (Si, Al)₃(O, N)₄:Eu), an α -sialon-based phosphor (e.g., Ca(Si, Al)₁₂(O, N)₁₆:Eu), or the like, nitride-based phosphors such as an LSN-based phosphor (e.g., (La, Y)₃Si₆N₁₁:Ce), a BSESN-based phosphor (e.g., (Ba, Sr)₂Si₅N₈:Eu), an SLA-based phosphor (e.g., SrLiAl₃N₄:Eu), a CASN-based phosphor (e.g., CaAlSiN₃:Eu), a SCASN-based phosphor (e.g., (Sr, Ca)AlSiN₃:Eu), or the like, fluoride-based phosphors such as a KSF-based phosphor (e.g., K₂SiF₆:Mn), a KSAF-based phosphor (e.g., K₂(Si_{1-x}Al_x)F_{6-x}:Mn, wherein x satisfies 0 < x < 1), a MGF-based phosphor (e.g., 3.5MgO.Math.0.5MgF₂.Math.GeO₂:Mn), or the like, a quantum dot having a perovskite structure (e.g., (Cs, FA, MA)(Pb, Sn)(F, Cl, Br, I)₃, wherein FA and MA are respectively formamidinium and methylammonium), a Group II-VI quantum dot (e.g., CdSe), a Group III-V quantum dot (e.g., InP), a quantum dot having a chalcopyrite structure (e.g., (Ag, Cu)(In, Ga)(S, Se)₂), etc.

[0068] The light-transmitting member **50** may include a filler such as a light-diffusing substance, etc. Examples of the light-diffusing substance include titanium oxide, barium titanate, aluminum oxide, YAP (yttrium aluminum perovskite), and silicon oxide. However, the light-diffusing substance is not limited thereto. The light-transmitting member **50** may include an optical film such

as an anti-reflection coating or the like on the upper surface, and may include an optical film such as a reflective film or the like at the lateral surface. The light-emitting device **1** may include the light-transmitting adhesive layer **51** such as a silicone resin or the like between the light-transmitting member **50** and the light-emitting element **20**.

[0069] When a voltage from the outside is applied via the anode and cathode pads of the light-emitting device **1**, a DC voltage is applied between the p-type semiconductor layer **26** and the n-type semiconductor layer **24** of the first semiconductor layer **22** via the pair of first electrodes **21** of the light-emitting element **20**; and the active layer **25** emits, for example, blue light. The blue light that is emitted from the active layer **25** passes through the n-type semiconductor layer **24**, the first element substrate **23**, and the bonding material layer **51** and is incident on the light-transmitting member **50**. The light-transmitting member **50** includes, for example, a phosphor that absorbs a portion of the incident blue light and emits yellow light. As a result, the light-emitting device **1** can emit white light in which the yellow light and the blue light from the light-transmitting member **50** are mixed. When a reverse surge current flows into the light-emitting device **1** due to static electricity, etc., the protection element **30** can protect the light-emitting element **20** from the surge current by carrying the surge current.

Effects

[0070] According to the embodiment, the process of removing the first element substrate **23** and the second element substrate **33** from the structure body **100** shown in FIG. 3 (hereinbelow, also called “the removal process”) includes removing a portion of the first element substrate **23** of the light-emitting element **20** and a portion of the second element substrate **33** of the protection element **30**. Therefore, the thickness of the light-emitting element **20** (specifically, the thickness of the first element substrate **23**) can be thinned. As a result, the amount of the light emitted from the first semiconductor layer **22** that is emitted laterally from the lateral surface of the first element substrate **23** can be reduced. As a result, the light extraction efficiency of the light-emitting device **1** is increased. Because the thickness of the protection element **30** (specifically, the thickness of the second element substrate **33**) also can be thinned, the light from the light-emitting element **20** that is absorbed by the protection element **30** can be reduced. The light extraction efficiency of the light-emitting device **1** is increased thereby.

[0071] According to the embodiment, the light-emitting element **20** and the protection element **30** are disposed on the wiring substrate **10** in the process of providing the structure body before the light-emitting element **20** and the protection element **30** are thinned in the removal process. It is therefore easy to handle the light-emitting element **20** and the protection element **30** in the process of disposing the light-emitting element **20** and the protection element **30** on the wiring substrate **10**. According to the embodiment, the thicknesses of the light-emitting element **20** and the protection element **30** after their portions are removed in the removal process are, for example, about 60 μm . If the light-emitting element **20** and the protection element **30** were to be pre-processed to such a thickness and then disposed on the wiring substrate **10**, there is a possibility that the light-emitting element **20** and the protection element **30** would be damaged by stress when disposing.

[0072] According to the embodiment, the first covering member **41** is disposed on the wiring substrate **10** in the process of disposing the first covering member. As a result, the light-emitting element **20** and the protection element **30** are fixed with respect to the wiring substrate **10**, and therefore the polishing is easy in the removal process.

[0073] According to the embodiment, it is sufficient for the first covering member **41** to cover at least the lateral surfaces of the first electrode **21** of the light-emitting element **20** and the wiring part **12** of the wiring substrate **10** and the lateral surfaces of the second electrode **31** of the protection element **30** and the wiring part **12** of the wiring substrate **10** in the process of disposing the first covering member **41**. As a result, detachment of the light-emitting element **20** and the protection element **30** from the wiring substrate **10** in the removal process can be reduced. In such a case, when the first electrodes **21** of the light-emitting element **20** and/or the second electrodes **31**

of the protection element **30** are connected to the wiring parts **12** of the wiring substrate **10** via a bonding member, the first covering member **41** covers the lateral surface of the bonding member. [0074] In the process of disposing the first covering member **41**, it is preferable for the first covering member **41** to cover the lateral surface of the light-emitting element **20** and the lateral surface of the protection element **30**; and it is more preferable for the first covering member **41** to collectively cover the upper surface of the light-emitting element **20** and the upper surface of the protection element **30** as shown in FIG. 2. As a result, damage such as cracking, chipping, etc., of the light-emitting element **20** and/or the protection element **30** in the first removal process can be reduced.

[0075] In the removal process according to the embodiment, only a portion of the second element substrate **33** of the protection element **30** is removed, and the entire second semiconductor layer **32** remains. As a result, the Zener voltage of the protection element **30** can be maintained.

[0076] In the process of disposing the second covering member according to the embodiment, the second covering member **42** is disposed on the first covering member **41** to cover the lateral surface **50c** of the light-transmitting member **50** but to leave the upper surface **50a** of the light-transmitting member **50** exposed. As a result, the light that is emitted from the lateral surface **50c** of the light-transmitting member **50** can be reflected by the second covering member **42** toward the light-transmitting member **50** and emitted from the upper surface **50a** of the light-transmitting member **50**. As a result, the light extraction efficiency is increased. By setting the reflectance of the second covering member **42** for the light emitted from the light-emitting element **20** to be greater than the reflectance of the first covering member **41** for the light emitted from the light-emitting element **20**, the light extraction efficiency is improved even further.

[0077] In the removal process according to the embodiment, a portion of the first element substrate **23**, a portion of the second element substrate **33**, and a portion of the first covering member **41** are removed by polishing the structure body **100** from the surface at the side opposite to the wiring substrate **10**. As a result, the upper surface **23a** of the first element substrate **23**, the upper surface **33a** of the second element substrate **33**, and the upper surface **41a** of the first covering member **41** can be positioned in the same plane **90**. As a result, the light emitted from the upper surface of the light-emitting element **20** of the light-emitting device **1** can be suppressed from being absorbed by the protection element **30** can be reduced, and the light-emitting device can have a high light extraction efficiency. Also, the light-emitting device **1** can be smaller.

Second Embodiment

[0078] The second embodiment differs from the first embodiment in that, after the first removal process, the remaining portion of the first covering member **41** is removed, and a covering member is newly disposed. For example, a thermoplastic polyimide resin is used as the first covering member **41**.

[0079] FIGS. 7 and 8 are cross-sectional views showing a method for manufacturing a light-emitting device according to the embodiment.

[0080] FIG. 9 is a cross-sectional view showing the light-emitting device according to the embodiment.

[0081] First, as shown in FIGS. 1 to 3, the process of providing the structure body, the process of disposing the first covering member, and the first removal process are performed.

Second Removal Process

[0082] Then, as shown in FIG. 7, the remaining portion of the first covering member **41** is removed. At this time, the wiring substrate **10**, the light-emitting element **20**, and the protection element **30** remain. For example, only the first covering member **41** is removed by dissolving with an organic solvent such as cyclopentanone, etc.

Process of Disposing Light-Transmitting Member 50

[0083] Continuing as shown in FIG. 8, the light-transmitting member **50** is disposed on the light-emitting element **20**. For example, the light-transmitting member **50** is bonded to the upper surface

23a of the first element substrate 23 of the light-emitting element 20 via the bonding material layer 51.

Process of Disposing Second Covering Member 42

[0084] Then, as shown in FIG. 9, the second covering member 42 is disposed on the wiring substrate 10. For example, the second covering member 42 can be disposed by potting, spraying, printing, transfer molding, injection molding, compression molding, etc. The second covering member 42 covers the upper surface 30a and the lateral surface 30b of the protection element 30, the lateral surface 20b of the light-emitting element 20, and the lateral surface 50c of the light-transmitting member 50, but does not cover the upper surface 50a of the light-transmitting member 50. The second covering member 42 may intrude between the wiring substrate 10 and the light-emitting element 20 and between the wiring substrate 10 and the protection element 30. For example, the second covering member 42 is formed of a resin material, e.g., a white resin material such as a silicone resin containing light-reflective particles. The second covering member 42 may be formed as a single layer or as multiple layers.

[0085] The second embodiment may include a process of disposing under-fill at the lower surface of the light-emitting element 20 and the upper surface of the wiring substrate 10 after the second removal process and before the process of disposing the second covering member 42. A resin material same as or similar to the second covering member 42 can be used as the under-fill.

[0086] Then, the wiring substrate 10 and the second covering member 42 are singulated into each set including one light-emitting element 20 and one protection element 30. Thus, the light-emitting device 2 according to the second embodiment is manufactured.

Light-Emitting Device 2

[0087] As shown in FIG. 9, the light-emitting device 2 according to the second embodiment includes the light-emitting element 20 and the protection element 30 located on the wiring substrate 10; and the upper surface of the light-emitting element 20 and the upper surface of the protection element 30 are positioned in the same plane. The light-emitting device 2 differs from the light-emitting device 1 shown in FIG. 5 in that the first covering member 41 that is positioned in the same plane as the upper surface of the light-emitting element 20 and the upper surface of the protection element 30 is not included. In the light-emitting device 2, the covering member 40 includes only the second covering member 42.

Effects

[0088] According to the second embodiment, the light-emitting element 20, the protection element 30, and the first covering member 41 are polished in the process shown in FIG. 3, then the remaining portion of the first covering member 41 is removed in the process shown in FIG. 7, and the second covering member 42 is deposited in the process shown in FIG. 9. As a result, the first covering member 41 does not remain in the light-emitting device 2 after manufacturing, and so the material of the first covering member 41 can be selected by considering only the ability to fix the light-emitting element 20 and the protection element 30 to the wiring substrate 10. For example, specifically, a hard resin material such as a polyimide resin can be used as the material of the first covering member 41. On the other hand, similarly to the first embodiment, a material selected with priority on the light reflectance and the heat resistance such as, for example, a white resin material such as a silicone resin can be used as the second covering member 42. Thus, suitable materials can be selected as the materials of the first and second covering members 41 and 42.

[0089] According to the second embodiment as well, it is preferable for the second covering member 42 to be formed of a material having a higher reflectance for the light emitted from the light-emitting element 20. Otherwise, the manufacturing method, the configuration, and the effects according to the second embodiment are the same as or similar to those of the first embodiment.

Third Embodiment

[0090] The configuration of the light-transmitting member according to the third embodiment is different from that of the first embodiment.

[0091] FIG. 10 is a cross-sectional view showing a light-emitting device according to the embodiment.

[0092] According to the embodiment as shown in FIG. 10, the light-emitting device 3 includes a light-transmitting member 52 instead of the light-transmitting member 50. The light-transmitting member 52 includes a resin layer 53 and a glass layer 54. The glass layer 54 is located on the resin layer 53. In the resin layer 53, for example, a phosphor 56 and/or a light-diffusing member 57 are located inside a base material 55 formed of a resin material such as silicone, etc. At least one of the phosphor 56 or the light-diffusing member 57 may not be included. The glass layer 54 may be formed of a light-transmitting material (e.g., a resin or a ceramic) other than glass.

[0093] According to the third embodiment, light of the desired color can be emitted by adjusting the type of the phosphor 56 and/or the type of the light-diffusing member 57 contained in the resin layer 53, etc. When manufacturing light-emitting devices that emit light of different colors, there are cases where the thickness of the resin layer 53 necessary for emitting the desired color is modified. However, according to the third embodiment, the light-emitting devices that emit light of different colors can be manufactured to have the same size by adjusting the thickness ratio of the resin layer 53 and the glass layer 54 in the light-transmitting member 52. Also, because the phosphor 56 is not exposed outside the light-emitting device 3, degradation of the phosphor 56 due to moisture in ambient air can be reduced. As a result, the reliability of the light-emitting device 3 is increased. Otherwise, the manufacturing method, the configuration, and the effects according to the third embodiment are the same as or similar to those of the first embodiment.

Fourth Embodiment

[0094] The fourth embodiment differs from the first embodiment in that the first element substrate and the second element substrate do not remain.

[0095] FIG. 11 is a cross-sectional view showing a method for manufacturing a light-emitting device according to the embodiment.

[0096] FIG. 12 is a cross-sectional view showing the light-emitting device according to the embodiment.

[0097] First, the processes shown in FIGS. 1 and 2 are performed.

[0098] Then, as shown in FIG. 11, the structure body 100 is polished from the surface at the side opposite to the wiring substrate 10. At this time, the entire first element substrate 23 of the light-emitting element 20 is removed, and a portion of the n-type semiconductor layer 24 of the first semiconductor layer 22 is removed. Also, the entire second element substrate 33 of the protection element 30 is removed, and a portion of the base material layer 34 of the second semiconductor layer 32 is removed.

[0099] Continuing, processes same as or similar to the processes shown in FIGS. 4 and 5 are performed.

[0100] As a result, as shown in FIG. 12, the light-emitting device 4 according to the fourth embodiment is manufactured. In the light-emitting device 4, the thicknesses of the light-emitting element 20, the protection element 30, and the first covering member 41 are, for example, about 20 μm .

[0101] As shown in FIG. 12, the light-emitting device 4 differs from the light-emitting device 1 according to the first embodiment in that the first element substrate 23 and the second element substrate 33 are not included, and the n-type semiconductor layer 24 and the base material layer 34 are thin. As a result, the light from the active layer 25 that is absorbed by the n-type semiconductor layer 24 is reduced; and the light that is emitted from the active layer 25 is not absorbed by the first element substrate 23. Furthermore, the light that is emitted from the active layer 25 is not absorbed by the second element substrate 33. As a result, the light extraction efficiency is improved even further. The light-emitting device 4 can be even smaller. Otherwise, the manufacturing method, the configuration, and the effects according to the embodiment are the same as or similar to those of the first embodiment.

Fifth Embodiment

[0102] The fifth embodiment is an example in which multiple pairs of the light-emitting element and the protection element are included in one light-emitting device.

[0103] FIG. **13** is a perspective view showing a light-emitting device according to the embodiment.

[0104] FIG. **14** is a cross-sectional view along line XIV-XIV shown in FIG. **13**.

[0105] FIG. **15** is a cross-sectional view along line XV-XV shown in FIG. **13**.

[0106] In the light-emitting device **5** according to the embodiment as shown in FIGS. **13** to **15**, multiple pairs of the light-emitting element **20** and the protection element **30** are located on one wiring substrate **10**. Multiple pads **13** that are used as external connection terminals of the light-emitting device **5** are located at the upper surface **10a** of the wiring substrate **10**.

[0107] In the light-emitting device **5**, for example, the multiple light-emitting elements **20** are periodically arranged in one column. According to the fifth embodiment, the arrangement direction of the light-emitting element **20** is taken as an “X-direction”. A direction that is parallel to the upper surface **10a** and orthogonal to the X-direction is taken as a “Y-direction”; and a direction that is orthogonal to the upper surface **10a** is taken as a “Z-direction”.

[0108] In the light-emitting device **5**, for example, the multiple protection elements **30** are also periodically arranged in one column along the X-direction. The number of the protection elements **30** is equal to the number of the light-emitting elements **20**. For example, the multiple pads **13** are also periodically arranged in one column along the X-direction. The number of the pads **13** is one greater than the number of the light-emitting elements **20**. In other words, when the number of the light-emitting elements **20** is n , the number of the protection elements **30** is n , and the number of the pads **13** is $(n+1)$. In the example shown in FIGS. **13** to **15**, n is 12.

[0109] In an example, the $(n+1)$ pads **13** arranged in one column connect the n light-emitting element **20** in series. The light-emitting device **5** can individually drive the light-emitting elements **20** by using the $(n+1)$ pads **13** as external connection terminals. The protection elements **30** are connected one-to-one with the light-emitting elements **20**. In the Y-direction, the column of the protection elements **30** is positioned between the column of the light-emitting elements **20** and the column of the pads **13**.

[0110] Similarly to the first embodiment, the light-emitting element **20** includes the first element substrate **23**. The first element substrate **23** is, for example, a sapphire substrate; and the fissility of sapphire can be utilized when singulating. Therefore, in the light-emitting element **20** after singulation, there are cases where the lateral surface of the first element substrate **23** includes surfaces that are not orthogonal to the upper surface **23a** due to the fissility of the first element substrate **23**. When the lateral surface of the first element substrate **23** is not orthogonal to the upper surface **23a**, for example, there are cases where the shape of the first element substrate **23** when viewed along the Y-direction is parallelogram-shaped instead of rectangular; and the position of the lower surface and the position of the upper surface **23a** of the first element substrate **23** are misaligned in the X-direction. Due to this misalignment, in the light-emitting device **5** that includes the multiple light-emitting elements **20**, there is a possibility that the distance between adjacent light-emitting elements **20** may vary.

[0111] In contrast, according to the embodiment, the first element substrate **23** is made thin in the process shown in FIG. **3**, and therefore the misalignment amount between the position of the lower surface and the position of the upper surface **23a** of the first element substrate **23** in the X-direction is reduced. As a result, the variation of the distances between the adjacent light-emitting elements **20** can be reduced.

[0112] For example, the light-emitting device **5** can be utilized as a light source of an automobile headlamp. For example, an ADB (adaptive driving beam) can be realized by independently controlling the multiple light-emitting elements **20**. Otherwise, the manufacturing method, the configuration, and the effects according to the fifth embodiment are the same as similar to those of the first embodiment.

TEST EXAMPLES

[0113] Test examples for the method for manufacturing the light-emitting device according to the first embodiment above will now be described in which the appearances of the upper surface **20a** of the light-emitting element **20**, the upper surface **30a** of the protection element **30**, and the upper surface **41a** of the first covering member **41** were observed, and the roughnesses were measured.

[0114] According to the test examples, multiple samples were provided, and the process of providing the structure body **100** shown in FIG. **1**, the process of disposing of the first covering member **41** shown in FIG. **2**, and the first removal process shown in FIG. **3** were performed. In the structure body **100**, the first support substrate was a sapphire substrate; and the second support substrate was a silicon substrate. In the disposition process, a silicone resin that included titanium oxide was disposed as the first covering member. To polish sapphire in the first removal process, a grindstone having a diamond abrasive and a grit of 6,000 was used to polish the upper surfaces of the samples. The upper surface **20a** of the light-emitting element **20**, the upper surface **30a** of the protection element **30**, and the upper surface **41a** of the first covering member **41** were simultaneously polished thereby. Then, the surfaces of the samples after polishing were observed using an optical microscope.

[0115] FIG. **16A**, FIG. **16B**, and FIGS. **17A** to **17C** are micrographs showing the appearances after polishing the light-emitting element, the protection element, and the first covering member.

[0116] FIGS. **16A** and **16B** were imaged using a surface roughness measuring device VK-3100 made by Keyence Corporation in a state in which laser light used for the roughness measurement was irradiated.

[0117] FIGS. **17A** to **17C** were imaged using a digital microscope VHX-8000 made by Keyence Corporation in a state in which light was irradiated by coaxial lighting in which the optical axis of the camera lens was substantially the same as the optical axis of the illumination light.

[0118] FIG. **16A**, FIG. **16B**, and FIGS. **17A** to **17C** are imaging results of mutually-different samples. In FIG. **16A**, FIG. **16B**, and FIGS. **17A** to **17C**, three line segments illustrate the direction of polishing marks resulting from the polishing.

[0119] As shown in FIG. **16A**, FIG. **16B**, and FIGS. **17A** to **17C**, polishing marks that extend in the same direction were formed in the upper surface **20a** of the light-emitting element **20** and the upper surface **30a** of the protection element **30**.

[0120] FIG. **18** is a table showing measurement results of the surface roughnesses of the light-emitting element, the protection element, and the first covering member.

[0121] As shown in FIG. **18**, the surface roughness (Sa) before and after polishing was measured by laser light irradiation using the surface roughness measuring device VK-3100 made by Keyence Corporation for the four samples. As a result, in the samples after polishing, the surface roughness (Sa) of the upper surface **30a** of the protection element **30** was greater and rougher than the surface roughness (Sa) of the upper surface **20a** of the light-emitting element **20**. The surface roughness (Sa) of the upper surface **41a** of the first covering member **41** was greater and rougher than the surface roughness (Sa) of the upper surface **30a** of the protection element **30**.

[0122] The sapphire of the upper surface **20a** of the light-emitting element **20** was harder than the silicon of the upper surface **30a** of the protection element **30**; and the silicon was harder than the silicone resin of the upper surface **41a** of the first covering member **41**. Therefore, by performing the polishing by using a grindstone of a grit that can polish sapphire, which was hardest, it is estimated that the surface roughness (Sa) could be higher for the materials having lower hardnesses.

[0123] FIG. **19** is a graph showing the surface profile of one sample. The horizontal axis is the horizontal-direction position; and the vertical axis is the vertical-direction position.

[0124] The surface profile was measured by laser light irradiation using the surface roughness measuring device VK-3100 made by Keyence Corporation for the one sample after polishing.

[0125] As shown in FIG. **19**, the upper surface **30a** of the protection element **30** and the upper

surface **20a** of the light-emitting element **20** were positioned in substantially the same plane. On the other hand, the upper surface **41a** of the first covering member **41** was positioned about 1 μm lower than the upper surface **30a** of the protection element **30** and the upper surface **20a** of the light-emitting element **20**. It is considered that this is because the first covering member **41** was softer than the protection element **30** and the light-emitting element **20** and easier to polish. [0126] The embodiments above are examples embodying the invention; and the invention is not limited to these embodiments. For example, additions, deletions, or modifications of some of the components or processes according to the embodiments above also are included in the invention. [0127] The invention can be utilized as, for example, a light source of a lighting device, etc.

Claims

1. A method for manufacturing a light-emitting device, the method comprising: providing a structure body, the structure body including a wiring substrate including a wiring part at an upper surface, a light-emitting element in which a pair of first electrodes, a first semiconductor layer, and a first element substrate are stacked, and a protection element in which a pair of second electrodes, a second semiconductor layer, and a second element substrate are stacked, wherein the pair of first electrodes and the pair of second electrodes face the upper surface of the wiring substrate, and the pair of first electrodes and the pair of second electrodes are connected to the wiring part; and removing at least a portion of the first element substrate of the light-emitting element and at least a portion of the second element substrate of the protection element from the structure body.
2. The method according to claim 1, further comprising between the providing of the structure body and the removing of the at least the portion of the first element substrate and the at least the portion of the second element substrate, disposing a first covering member on the wiring substrate, wherein the first covering member fixes the first electrodes of the light-emitting element, the second electrodes of the protection element, and the wiring substrate.
3. The method according to claim 2, wherein the disposing of the first covering member includes covering the first element substrate and the second element substrate with the first covering member, and the removing of the at least the portion of the first element substrate and the at least the portion of the second element substrate includes a first removal process of removing at least the portion of the first element substrate, at least the portion of the second element substrate, and a portion of the first covering member from a side of the structure body opposite to the wiring substrate.
4. The method according to claim 3, wherein the removing of the at least the portion of the first element substrate and the at least the portion of the second element substrate further includes a second removal process of removing a remaining portion of the first covering member after the first removal process.
5. The method according to claim 1, wherein the first semiconductor layer includes an n-type semiconductor layer, an active layer, and a p-type semiconductor layer in this order from the first element substrate side, and the removing of the at least the portion of the first element substrate and the at least the portion of the second element substrate includes removing an entirety of the first element substrate and removing a portion of the n-type semiconductor layer.
6. The method according to claim 1, wherein the second element substrate is a semiconductor substrate, the second semiconductor layer includes a first semiconductor region connected to one of the second electrodes, and a second semiconductor region spaced apart from the first semiconductor region, the second semiconductor region being connected to the other of the second electrodes, and the removing of the at least the portion of the first element substrate and the at least the portion of the second element substrate includes removing the portion of the second element substrate while allowing an entirety of the second semiconductor layer to remain.
7. The method according to claim 2, further comprising after the removing of the at least the

portion of the first element substrate and the at least the portion of the second element substrate: disposing a light-transmitting member on the light-emitting element; and disposing a second covering member on the first covering member, the second covering member covering a lateral surface of the light-transmitting member, and the second covering member not covering an upper surface of the light-transmitting member.

8. The method according to claim 4, further comprising after the second removal process: disposing a light-transmitting member on the light-emitting element; and disposing a second covering member on the wiring substrate, the second covering member covering a lateral surface of the protection element, a lateral surface of the light-emitting element, and a lateral surface of the light-transmitting member, while not covering an upper surface of the light-transmitting member.

9. The method according to claim 3, wherein the first removal process includes polishing the structure body from a surface at the side opposite to the wiring substrate.

10. A light-emitting device comprising: a wiring substrate; a light-emitting element arranged on the wiring substrate; a protection element arranged on the wiring substrate; a light-transmitting member arranged on the light-emitting element; and a covering member arranged on the wiring substrate, the covering member not covering an upper surface of the light-transmitting member while covering a lateral surface of the light-emitting element and a lateral surface of the protection element, wherein an upper surface of the light-emitting element and an upper surface of the protection element are positioned in a same plane.

11. The light-emitting device according to claim 10, wherein the covering member includes a first covering member covering the lateral surface of the light-emitting element and the lateral surface of the protection element, an upper surface of the first covering member being positioned in the same plane as the upper surface of the light-emitting element and the upper surface of the protection element; and a second covering member arranged on the first covering member, the second covering member covering the upper surface of the protection element and a lateral surface of the light-transmitting member.

12. The light-emitting device according to claim 11, wherein the second covering member has a higher reflectance than the first covering member for light emitted from the light-emitting element.

13. The light-emitting device according to claim 10, wherein a thickness of the light-emitting element is not less than 20 μm and not more than 80 μm .

14. The light-emitting device according to claim 10, wherein a surface roughness of the upper surface of the protection element is greater than a surface roughness of the upper surface of the light-emitting element.

15. The light-emitting device according to claim 10, wherein a surface roughness of the upper surface of the first covering member is greater than a surface roughness of the upper surface of the light-emitting element and a surface roughness of the upper surface of the protection element.
